Section 1: Probabilities, Distributions, and Random Variables

A **random experiment** is a procedure where every possible outcome can be described prior to its performance and is also repeatable

Example 1.1:

Roll a dice

All possible outcomes: $\{1, 2, \ldots, 6\}$

Notation:

$\mathcal{C}$: sample space

C : collection of elements/events/outcomes that belong to $\mathcal{C}$, e.g., C = odd values

$\dot{C}$: specific outcome of the experiment, e.g., $\dot{C} = 1$

$\Rightarrow \dot{C} \in C \in \mathcal{C}$

$\Rightarrow \dot{C} \subset C \subset \mathcal{C}$

we can count how many times each event/outcome $\dot{C}$ occurs in N trials, which gives us our frequency (f), which allows us to calculate a relative frequency (f/N), which is the basis of probabilities

Example 1.2: In the game of craps, we are not interested in the specific outcomes, but a function of the outcomes, i.e. the sum of the two dice where $D_i = \{1, 2, 3, 4, 5, 6\}$. The total number of outcomes is 36 and to win, we want a 7 or an 11, the former of which has a 6/36 chance of occurring. What about 11?

Notation:

$$C = \{(1,1), (1,2), \ldots, (6,5), (6,6)\}$$ 36 possible combinations

$$C = \{(5,6), (6,5)\} \Rightarrow Pr(c) = \frac{2}{36}$$

$$c = (5,6)$$

$$C = \{x : 0 \le x \le 1\}$$ 1-D set of points for x → notation for a continuous RV

$$C = \{(x,y) : 0 \le x \le 1, 0 \le y \le 1\}$$ 2-D set of points (x,y) that form all points within a square

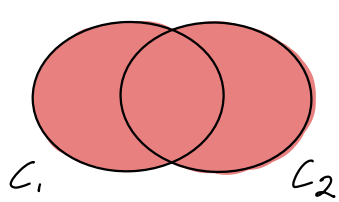
Definition 1.2.1: If each element of a set C_1 is also an element of C_2, then the set of C_1 is a **subset** of C_2, i.e. $C_1 \subset C_2$.

C_1 = odd values, C_2 = all values

$$\Rightarrow C_1 \subset C_2$$

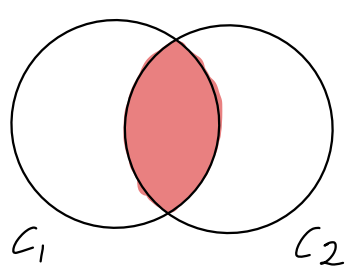
Definition 1.2.2: If each element of a set C has no elements, then C is a considered a **null subset**, where $C = \emptyset$, also known as an **empty set**.

Definition 1.2.3: The set of all elements that belong to *at least one* of the sets C_1 and C_2 is called the **union** of C_1 and C_2, i.e. $C_1 \cup C_2$.



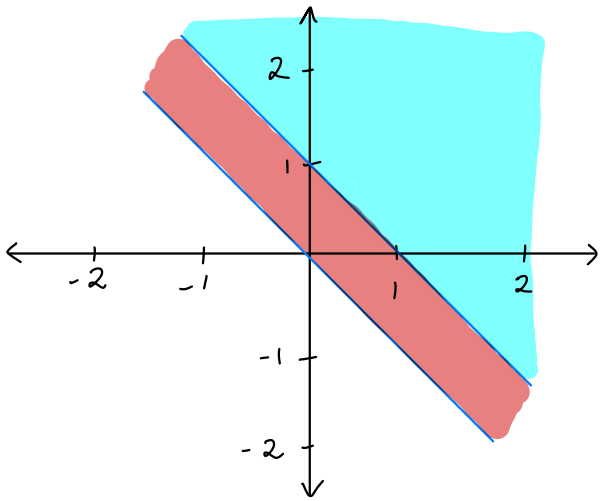
: $C_1 \cup C_2$

Definition 1.2.4: The set of all elements that belong to *both* C_1 and C_2 is called the **intersection** of C_1 and C_2, i.e. $C_1 \cap C_2$.



: $C_1 \cap C_2$

Example 1.3: Let $C_1 = \{(x, y): 0 \leq x + y \leq 1\}$ and $C_2 = \{(x, y): 1 < x + y\}$, what is $C_1 \cap C_2$?



$$C_1 \cap C_2 = \emptyset$$

C_1 and C_2 share no points, i.e., nothing is in common

IF $C_2 = \{(x, y): 1 \leq x + y\}$

$\Rightarrow C_1 \cap C_2 = \{(x, y): x + y = 1\}$

Definition 1.2.5: The set of all elements under consideration is called the **sample space** $\mathcal{C}$.

Definition 1.2.6: Let $\mathcal{C}$ be the space and C be a subset of $\mathcal{C}$ such that $(\exists)\ C \subset \mathcal{C}$, then the set of all elements *not* in C is the complement of C: C^c or C'

Note: if C contains all elements, then $C^c = \emptyset$
in other words, $\emptyset^c = \mathcal{C}$

DeMorgan, Commutative, Associative, and Distributive Laws:

De Morgan: $(C_1 \cap C_2)^c = C_1^c \cup C_2^c$

Commutative: $C_1 \cup C_2 = C_2 \cup C_1$

$C_1 \cap C_2 = C_2 \cap C_1$

Associative: $(C_1 \cup C_2) \cup C_3 = C_1 \cup (C_2 \cup C_3)$

$(C_1 \cap C_2) \cap C_3 = C_1 \cap (C_2 \cap C_3)$

Distributive: $C_1 \cup (C_2 \cap C_3) = (C_1 \cup C_2) \cap (C_1 \cap C_3)$

$C_1 \cap (C_2 \cup C_3) = (C_1 \cap C_2) \cup (C_1 \cap C_3)$

Probabilities and Set Functions

Definition 1.3.1: Let $\mathcal{B}$ be a set of events, then $P(\mathcal{B})$ must satisfy the following conditions:

1. $P(\mathcal{B}) \geq 0$
2. $P(\mathcal{B}) \leq 1$
3. $P(\mathcal{C}) = 1$
4. If $\{C_n\}$ is a sequence of events in $\mathcal{B}$ and $\{C_m \cap C_n\} = \emptyset$, then $\forall\, m \neq n$

$$P(\cup\, C_i\} = \sum_{i \in m,n} P(C_i)$$

This may look weird, but it is familiar, as this is a more generic description of the additive rule for mutually exclusive / disjoint events

$$Pr(A \cup B) = Pr(A) + Pr(B), \text{ where } A \cap B = \emptyset$$

where C_m, C_n are ME$_x$ / disjoint $\Rightarrow \{C_m \cap C_n\} = \emptyset$

A collection of events whose members are pairwise disjoint, as in (4), is said to be **mutually exclusive**

Theorem 1.3.1: $\forall\, C \in \mathcal{B}, P(C) = 1 - P(C^c)$

Proof: by defn. 1.2.6: $C \cup C^c = \mathcal{C}$, $C \cap C^c = \emptyset$

then, by defn 1.3.1 (2): $Pr(\mathcal{C}) = Pr(C \cup C^c) = 1$

$$Pr(C) + Pr(C^c) = 1$$

$$Pr(C) = 1 - Pr(C^c)$$

(or) $Pr(C) = 1 - Pr(C)$

Theorem 1.3.2: The probability of the null set is 0, i.e. $P(\emptyset) = 0$.

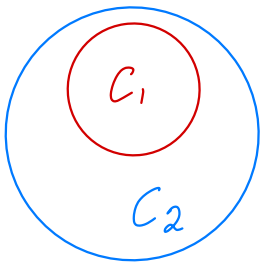
Proof: let $C = \emptyset \Rightarrow C^c = \mathcal{C}$, then by theorem 1.3.1,

$$Pr(C) = Pr(\emptyset) = 1 - Pr(\mathcal{C})$$

$$= 1 - 1$$

$$= 0$$

Theorem 1.3.3: If C_1 and C_2 are events $\ni C_1 \subset C_2$, then the $P(C_1) \leq P(C_2)$



Proof: $C_2 = C_1 \cup (C_1^c \cap C_2)$, $C_1 \cap (C_1^c \cap C_2) = \emptyset$

By defn. 1.3.1, $Pr(C_2) = Pr(C_1) + Pr(C_1^c \cap C_2)$

By defn 1.3.1 (2), $Pr(C_1^c \cap C_2) \geq 0$

$\Rightarrow Pr(C_2) \geq Pr(C_1)$

Theorem 1.3.4: For each $C \in \mathcal{B}$, $0 \leq P(C) \leq 1$

Proof: By defn. 1.3.1, the conditions for $Pr(B)$ hold for $Pr(C)$ since $C \in B$

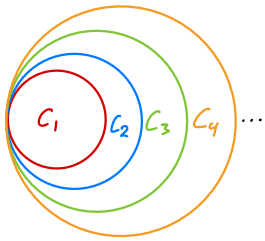
$\Rightarrow \exists\, C \in B$ s.t. $Pr(C) \leq 1$, $Pr(C) \geq 0$

Theorem 1.3.5: If C_1 and C_2 are events in $\mathcal{C}$, then $P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$

↳ Additive Rule

A sequence of events $\{C_n\}$ is a $\underbrace{\textbf{non-decreasing}}$ **sequence** if $C_n \subset C_{n+1}\ \forall\, n$

↳ monotone increasing



$C_1 \subset C_2 \subset \dots$, $C_1 \subseteq C_2 \subseteq \cdots$, or something in between

$\Rightarrow \lim_{n\to\infty} C_n = \bigcup_{n=1}^{\infty} C_n$

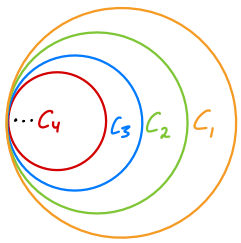
Theorem 1.3.6: Let $\{C_n\}$ be a *non-decreasing* sequence of events, then

$$\lim_{n\to\infty} P(C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

Follows from above

and if $\{C_n\}$ is a *decreasing* sequence of events, then

$$\lim_{n\to\infty} P(C_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right)$$

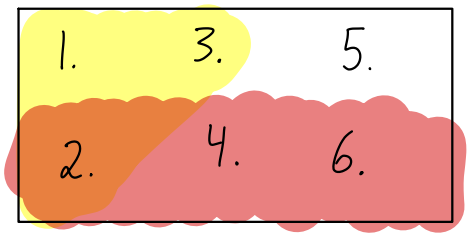


$\cdots \subset C_4 \subset C_3 \subset C_2 \subset C_1$

Conditional Probabilities and Independence

Let $C_1, C_2 \subset \mathcal{C}$, then the **conditional probability** of event C_2 relative C_1, i.e. the conditional probability of C_2 given C_1 is denoted as $P(C_2 \mid C_1)$. Importantly, $\mathcal{C}$ is *no longer our sample space*, but instead C_1. We define this conditional probability as:

let X be a dice roll, A = even, B = roll < 4



$$Pr(A) = \frac{3}{6} = \frac{1}{2}$$

$$Pr(A|B) = \frac{1}{3}$$

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

$$Pr(C_2|C_1) = \frac{Pr(C_2 \cap C_1)}{Pr(C_1)} \quad \text{where } Pr(C_1) > 0$$

The following conditions (still) hold true:

1. $P(C_2 \mid C_1) \geq 0$ and $P(C_2|C_1) \leq 1$
2. $\forall\, C_j, C_i$ that are mutually exclusive (MEx)/disjoint,

$$P\left(\bigcup_{j=2}^{\infty} C_j \mid C_i\right) = \sum P(C_j \mid C_i)$$

3. $P(C_i \mid C_i) = 1 \;\forall\, i$

The **multiplication rule** uses our definition of conditional probability and rearranges so that the probability of an intersection between two events can be defined as the product between a conditional probability and what we are conditioning on:

$$P(C_1 \cap C_2) = P(C_2|C_1) \cdot P(C_1) = P(C_1|C_2) \cdot P(C_2)$$

which can be extended to an n sequence of events.

Example 1.4:

if we have a standard deck of cards w/ 52 values

let A = Ace

$Pr(A) = \frac{4}{52} = \frac{1}{13}$

if we draw an ace on the first card, what is the probability of getting an ace on the second draw?

$Pr(A_2|A_1) = \frac{3}{51}$

$\Rightarrow Pr(A_1 \cap A_2) = Pr(A_1) \cdot Pr(A_2|A_1)$

$= \frac{4}{52} \cdot \frac{3}{51}$

The **law of total probability** takes advantage of the above, as we can define the probability of some event C as:

$$P(C) = \sum P(C_i) \cdot P(C|C_i)$$

where $C_1, C_2, \ldots$ are mutually exclusive/disjoint events

Bayes Theorem, which you all saw in STAT101/224/231 and STAT333, takes advantage of the law of total probability, conditional probabilities, and the multiplication rule to understand the likelihood of one conditional probability based on changing the direction of the conditional statement,

$$P(C_j|C) = \frac{P(C \cap C_j)}{P(C)} = \frac{P(C_j) \cdot P(C\ |C_j)}{P(C)}$$

$$= \frac{P(C_j) \cdot P(C\ |C_j)}{\sum P(C_i) \cdot P(C|C_i)}$$

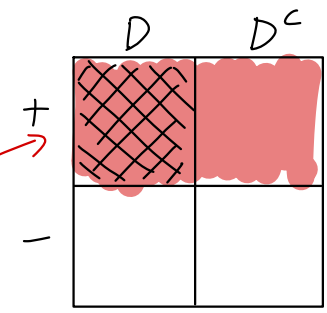
where $i, j \in (1, 2, \dots, n)$ and i, j are mutually exclusive

Example 1.5: Suppose a disease is prevalent in a population with a rate of 0.5% and a rapid blood test has an accuracy of 99% among those with the disease, i.e. returns a positive 99% if the disease is present, and a false positive 5% of the time among those without the disease, i.e. returns a positive 5% if the disease is not present.

$Pr(D) = 0.005 \Rightarrow Pr(D^c) = 0.995$

$Pr(+|D) = 0.99$

$Pr(+|D^c) = 0.05$



$Pr(+) = Pr(+ \cap D) + Pr(+ \cap D^c)$

$$Pr(D|+) = \frac{Pr(D \cap +)}{Pr(+)}$$

$$= \frac{Pr(+|D) \cdot Pr(D)}{Pr(+|D) \cdot Pr(D) + Pr(+|D^c) \cdot Pr(D^c)}$$

$$= \frac{0.99 \cdot 0.005}{0.99 \cdot 0.005 + 0.05 \cdot 0.995}$$

$$= 0.905 = 9.05\%$$

If C_1 has *no effect* on C_2, and vice-versa, then $P(C_2|C_1) = P(C_2)$ and $P(C_1|C_2) = P(C_1)$, and we say that C_1 and C_2 are **independent** of one another

Definition 1.4.1: Let C_1 and C_2 be two events, we say C_1 and C_2 are IND if and only if (IFF, $\Leftrightarrow$)

$$P(C_1 \cap C_2) = P(C_1) \cdot P(C_2)$$

Properties: If C_1 and C_2 are IND

1. C_1 and C_2^c are IND
2. C_1^c and C_2 are IND
3. C_1^c and C_2^c are IND

Be able to prove these

For the purposes of this course, you can assume that **if events are IND, they are also mutually independent**, where

$$P(C_1 \cap C_2 \cap \ldots \cap C_n) = P(C_1) \cdot P(C_2) \cdot \ldots \cdot P(C_n) = \prod_{i=1}^{n} P(C_i)$$

this idea and notation will be visited again

Random Variables

Definition 1.5.1: Consider a random experiment with a sample space $\mathcal{C}$. A function X, which assigns to each element $c \in \mathcal{C}$ one and only one value, this is called a **random variable**

Example 1.6: $\mathcal{C} = \{H, T\}$, X be a function $\Rightarrow X(H) = 1,\ X(T) = 0$

$$X(c) = \begin{cases} 1, & c = H \\ 0, & c = T \end{cases}$$

Notation:

$p(x) = PMF \Rightarrow x \sim$ discrete RV

$f(x) = PDF \Rightarrow x \sim$ continuous RV

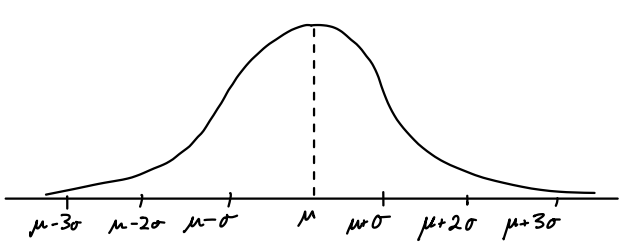
Definition 1.5.2: Let X be a random variable, its **cumulative distribution function (CDF)** is defined by $F_X(x)$ where

$$F_X(x) = P(X \le x) = \begin{cases} \sum_x p_X(x), & X \text{ is discrete} \\ \int_{-\infty}^{x} f_X(x)\ dx, & X \text{ is continuous} \end{cases}$$

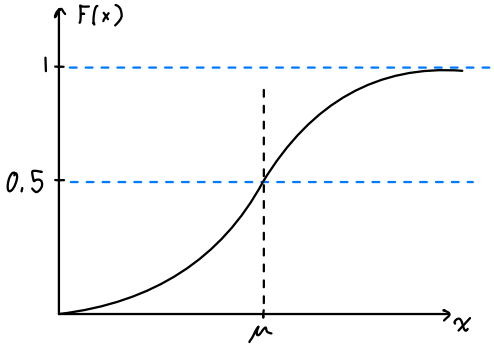
Theorem 1.5.1: Let X be a random variable with CDF $F(x)$, then

1. $\forall\ a, b \in \mathcal{C}$, if $a \le b \Rightarrow F(a) \le F(b)$
2. $\lim_{x \to -\infty} F(x) = 0$
3. $\lim_{x \to \infty} F(x) = 1$
4. $\lim_{x \downarrow x_0} F(x) = F(x_0) \Rightarrow F$ is right-side continuous

For points (2) and (3), consider the normal distribution where the pdf: $N(\mu, \sigma)$ → always the std. dev.



since the lower bound is $-\infty$ and lim ? is $-\infty$, then $\lim_{x \to -\infty} F(x) = 0$



why the sigmodial shape? As we move to the right of $\mu \Rightarrow \lim f(x) \downarrow$ so how much we add-on to $F(x)$ gets "smaller", hence the commulative growth slows down since $F(x) = Pr(X \leq x)$

Theorem 1.5.2: Let X be a random variable, then $\forall\, a < b, P(a < X \leq b) = F_X(b) - F_X(a)$.

$$
\begin{aligned}
Pr(a < x \leq b) &= Pr(a < x \cap x \leq b) \\
&= Pr(x \leq b) + Pr(x > a) - Pr(a < x \cup x \leq b) \\
&= Pr(x \leq b) + Pr(x > a) - 1 \\
&= Pr(x \leq b) + 1 - Pr(x \leq a) - 1 \\
&= Pr(x \leq b) - Pr(x \leq a) \\
&= F_x(b) - F_x(a)
\end{aligned}
$$

For some more basic details and properties, please review sections 1.6 – 1.7 of the Hogg text

Transformations: Discrete and Continuous Random Variables

We have a random variable X and know its distribution, but we are interested in Y, which is some transformation (i.e. function) of $X \ni Y = g(X)$.

Assume $X \sim$ discrete random variable, then the space of Y $\mathcal{D}_Y = \{g(x): x \in \mathcal{D}_X\}$ and the transformation of X to Y is one-to-one, then,

$$p_Y(y) = P(Y = y) = P(g(X) = y) = P\big(X = g^{-1}(y)\big) = p_X(g^{-1}(y))$$

Example 1.7:

Let X be the number of flips to get the first head from a coin flip

$\Rightarrow p_X(x) = Pr(X = x) = \left(\frac{1}{2}\right)^x$, where $X = 1, 2, \ldots$

$\mathcal{D}_X = \{1, 2, \ldots\}$

Let Y be the number of flips before getting the first head from a coin flip

$\Rightarrow Y = g(x) = x - 1$

$\Rightarrow g^{-1}(y) = y + 1$, $\mathcal{D}_y = \{0, 1, \ldots\}$

$$\Rightarrow p_y(y) = p_x(g^{-1}(y))$$

$$= \left(\frac{1}{2}\right)^x \Big|_{x = y+1}$$

$$= \left(\frac{1}{2}\right)^{y+1}, \quad \mathcal{D}_y = \{0, 1, \ldots\}$$

When $X \sim$ continuous random variable, the transformation to Y is not as straightforward and we must obtain the PDF through the CDF:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P\big(X \leq g^{-1}(y)\big) = F_X(g^{-1}(y))$$

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Theorem 1.7.1: Let X be a continuous random variable w PDF $f_X(x)$ and a support $\mathcal{S}_X$, where the **support** is defined as the probability of X is non-trivial. Let $Y = g(X)$ be a one-to-one differentiable function, then the PDF of Y is:

$$f_Y(y) = \frac{d}{dy} F_X\big(g^{-1}(y)\big) = f_X\big(g^{-1}(y)\big) \cdot \left|\frac{dx}{dy}\right|, y \in \mathcal{S}_Y$$

where $\mathcal{S}_Y = \{y = g(x) : x \in \mathcal{S}_X\}$

Notation: $\left|\frac{dx}{dy}\right| = \left|\frac{d}{dy} x\right| = \left|\frac{d}{dy} g^{-1}(y)\right| = |J|$

Example 1.8:

support (S_x)

always assume base e

Let $f(x) = \begin{cases} 1, & 0 < x \leq 1 \\ 0, & e/w \end{cases} \Rightarrow x \sim Unif(0,1)$

let $Y = -2\log(x) = -2\ln(x)$

$\Rightarrow g(x) = -2\log(x)$

$\Rightarrow g^{-1}(y) = e^{-\frac{1}{2}y}$

$\Rightarrow f_Y(y) = f_X(g^{-1}(y)) \cdot |J|$

$= f_X\left(e^{-\frac{1}{2}y}\right) \cdot \frac{1}{2} e^{-\frac{1}{2}y}$

$= \frac{1}{2} e^{-\frac{1}{2}y}$

Going back to 333, you may have seen the following

$$F_Y(y) = Pr(Y \le y) = Pr(-2\log(X) \le y) = Pr(X \ge e^{-\frac{y}{2}})$$

Be cautious of inequalities and negative values

$$= \int_{e^{-\frac{y}{2}}}^{1} f(x)\,dx = \int_{e^{-\frac{y}{2}}}^{1} 1\,dx = x\Big|_{x=e^{-\frac{y}{2}}}^{x=1} = 1 - e^{-\frac{y}{2}}$$

Recall: since $X \sim Unif(0,1) \Rightarrow$ the support is between 0 and 1 $\Rightarrow$ the upper bound for $Pr(X \ge e^{-y/2})$ must be 1

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy}\left(1 - e^{-\frac{y}{2}}\right) = \frac{1}{2} e^{-\frac{y}{2}}$$

Q: What is the support? i.e. the space of Y

↳ Where the probability is positive / non trivial

$$S_x = \{x : 0 < x \le 1\}$$

$$S_y = \{y = -2\log(x) : a < y < b\}$$

$y = -2\log(x) = \log(x^{-2}) = \log(\frac{1}{x^2})$ where x is restricted by its S_x

$$\lim_{x \to 1} \log\left(\frac{1}{x^2}\right) = 0, \quad \lim_{x \to 0} \log\left(\frac{1}{x^2}\right) = +\infty$$

$$\Rightarrow S_y = \{y : 0 < y < +\infty\}$$

$$\therefore f_Y(y) = \begin{cases} \frac{1}{2} e^{-\frac{1}{2}y}, & 0 < y < \infty \\ 0, & e/w \end{cases}$$

The most obvious (and useful) transformation (for those that did STAT331 or 357) is the generalized linear model, which is based on our classic straight-line. If we recall STAT231, we start with the line of best fit from high school:

$$y = mx + b$$

and update the notation

Simple Linear model: $y = \beta_0 + \beta_1 x$

where y is continuous; however, if y isn't continuous or when our assumption of normality doesn't work, and the idea of a transformation becomes very important as it allows us to use a similar methodology.

For those who have taken/are taking Statistical Methods (331) or Regression (357), you'll see something called a **linear predictor**, which is the right-hand side of the equation used repeatedly, regardless of the outcome measure.

Case in point, in epidemiology we often have a binary outcome (D vs. D^c), which clearly doesn't follow a continuous distribution or allow for the normality assumption. Using a transformation, we can maintain the same linear predictor above:

$$\text{outcome: } y \rightarrow logit(y) = log\left(\frac{y}{1-y}\right)$$

We run into one small issue, though, since y is either D or D^c, which is often 1 or 0 (binary), this would cause the equation above to blow up. We can make a minor modification and instead let y be replaced by p, the probability of disease, which we can dichotomize into either D or D^c.

$$\text{outcome: } y \rightarrow logit(p) = log\left(\frac{p}{1-p}\right), \text{ where } p = Pr(D)$$

so that an equivalent equation of our simple linear model becomes the

Simple logistic model: $logit(p) = log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$

Expectations: Discrete and Continuous Random Variables

Definition 1.8.1: Let X be some random variable, if $X \sim$ continuous random variable with PDF $f(x) \ni$

$$\int_{-\infty}^{\infty} |x| \cdot f(x)\, dx < \infty \Big\} \text{ i.e., it exists/is defined}$$

then the **expectation of X** is:

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

and if $X \sim$ discrete random variable with PMF $p(x) \ni$

$$\sum |x| \cdot p(x) < \infty$$

then the **expectation of X** is:

$$E(X) = \sum x \cdot p(x)$$

Definition 1.9.1: Let X be a random variable $\ni E(X) \exists$, the **mean value μ of X** is defined as $E(X) = \mu$

Regardless of the type of random variable, the expectation is also known as the **first moment**, which we will touch on in a moment.

Example 1.9: WLOG, let's assume that a random variable follows a continuous distribution $f(x)$ and c is some constant, then:

$$E(cX) = \int_{-\infty}^{\infty} c x \cdot f(x)\, dx = c \cdot \int_{-\infty}^{\infty} x \cdot f(x)\, dx = c \cdot E(X), \text{ where } c \in \mathbb{R}$$

As we can see, the impact of a constant c is minimal and simply scales the mean μ.

Theorem 1.8.2: Let $g_1(x)$ and $g_2(x)$ be functions of a random variable X, then suppose $E(g_1(x))$ and $E\big(g_2(x)\big)$ exist, then for $k_1, k_2 \in \mathbb{R}$

$$E\big(k_1 \cdot g_1(x) + k_2 \cdot g_2(x)\big) = k_1 \cdot E\big(g_1(x)\big) + k_2 \cdot E\big(g_2(x)\big)$$

Definition 1.9.2: Let X be a random variable with a finite mean $\mu \ni E([X-\mu]^2)$ is finite, then the **variance of $\boldsymbol{X}$, $\boldsymbol{\sigma^2}$** or $\mathrm{Var}(X)$, is $E([X-\mu]^2)$

As we established previously, there is a close relationship between the mean and variance of X, where we can define the variance based on the difference of the second moment and the square of the first moment.

$$\sigma^2 = E(X^2) - E(X)^2$$

This format should feel/look somewhat familiar if we consider the formula for the sample variance (s^2) from STAT101/231:

$$s^2 = \frac{1}{n-1} \cdot \sum (x_i - \bar{x})^2 \qquad \underline{\text{or}} \qquad s^2 = \frac{1}{n-1}\left[\sum (x_i^2) - \frac{\sum (x_i)^2}{n}\right]$$

We can determine any n^{th} moment by the equation:

$$E(X^n) = \begin{cases} \sum x^n \cdot p(x)\,, & X \sim \text{discrete} \\ \int x^n \cdot f(x)\,dx\,, & X \sim \text{continuous} \end{cases}$$

however, as mentioned before, there are multiple ways we can approach problems, and another way that can be easier for us to determine these moments is through a generic approach to moments.

With that said, try (or confirm) that for $X \sim$ cont. RV

$\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$, where $a, b \in \mathbb{R}$

Definition 1.9.3: Let X be a random variable $\ni \exists\ h > 0$ and $E(e^{tX})$ for $-h < t < h$, then the **moment generating function (MGF) of X** is defined as:

$$M(t) = E(e^{tX}) = \begin{cases} \sum e^{tx} \cdot p(x)\,, & X \sim \text{discrete} \\ \int e^{tx} \cdot f(x)\,dx\,, & X \sim \text{continuous} \end{cases}$$

As such, to calculate our moment(s) we take the derivative with respect to t like so:

$$M'(t) = \begin{cases} \frac{d}{dt}M(t) = \frac{d}{dt}\sum e^{tx}\cdot p(x) = \frac{d}{dt}\sum e^{xt}\cdot p(x), & X \sim \text{discrete RV} \\ \frac{d}{dt}M(t) = \frac{d}{dt}\int e^{tx}\cdot f(x)\,dx = \frac{d}{dt}\int e^{xt}\cdot f(x)\,dx, & X \sim \text{continuous RV} \end{cases}$$

and we can then evaluate $M'(t = 0)$:

$$M'(t=0) = E(X) = \begin{cases} \sum xe^{xt}\cdot p(x)\Big|_{t=0} = \sum x\cdot p(x), & X \sim \text{discrete RV} \\ \int xe^{xt}\cdot f(x)\,dx\Big|_{t=0} = \int x\cdot f(x)\,dx, & X \sim \text{continuous RV} \end{cases}$$

Taking the second derivative of t and evaluating it at 0 produces another familiar looking equation

$$M''(t) = \begin{cases} \frac{d}{dt}\left[\sum xe^{xt}\cdot p(x)\right] = \sum x^2e^{xt}\cdot p(x), & X \sim \text{discrete RV} \\ \frac{d}{dt}\left[\int xe^{xt}\cdot f(x)\,dx\right] = \int x^2e^{xt}\cdot f(x)\,dx, & X \sim \text{continuous RV} \end{cases}$$

$$M''(t=0) = \begin{cases} \sum x^2\cdot p(x), & X \sim \text{discrete RV} \\ \int x^2\cdot f(x)\,dx, & X \sim \text{continuous RV} \end{cases}$$

In general, if m is a positive integer ($m \in \mathbb{I}^+$) and if $M^m(t=0) = E(X^m)$, we have defined the moments of X.

$X \sim Exp(\lambda = 1)$ ($\lambda e^{-\lambda x}$ in general)

Example 1.10:

Let $X \sim$ continuous RV where $F(x) = \begin{cases} e^{-x}, & 0 < x < \infty \\ 0, & e/w \end{cases}$

$$M(t) = E(e^{tx}) = \int_0^\infty e^{tx} \cdot e^{-x}\, dx$$
$$= \int_0^\infty e^{tx-x}\, dx$$
$$= \int_0^\infty e^{x(t-1)}\, dx$$
$$= \left(\frac{1}{t-1}\right) e^{x(t-1)} \Big|_0^\infty = \frac{1}{1-t} = (1-t)^{-1}$$

$$E(X) = M'(t=0)$$
$$= \frac{d}{dt}\left[(1-t)^{-1}\right]\Big|_{t=0}$$
$$= \left[(-1)(1-t)^{-2} \frac{d}{dt}(1-t)\right]\Big|_{t=0}$$
$$= (1-t)^{-2}\Big|_{t=0}$$
$$= 1 = \lambda = \mu = E(X)$$

Some common MGFs:

Binomial:	$X \sim bin(n,p)$	$(p \cdot e^t + [1-p])^n$
Poisson:	$X \sim Pois(\lambda)$	$\exp(\lambda \cdot [e^t - 1])$
Uniform:	$X \sim Unif(a,b)$	$\dfrac{e^{tb} - e^{ta}}{t(b-a)}$
Exponential:	$X \sim Exp(\lambda)$	$\dfrac{\lambda}{\lambda - t}$
Normal:	$X \sim N(\mu,\sigma)$	$\exp\left(\mu t + \dfrac{\sigma^2 t^2}{2}\right)$

Example 1.11:

$X \sim Pois(\lambda) \quad E(x) = \lambda = V(x) \quad E(x^2) - E(x)^2 \Rightarrow E(x^2) = \lambda^2 + \lambda$

$$M(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx}\left[e^{-\lambda} \cdot \frac{\lambda^x}{x!}\right]$$
$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$$
$$= e^{-\lambda}\left(e^{\lambda e^t}\right)$$
$$= e^{\lambda(e^t - 1)}$$
$$= \exp(\lambda(e^t - 1))$$

Recall: $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
$= \sum_{n=0}^{\infty} \frac{x^n}{n!}$ } power series

$M'(t) = \frac{d}{dt}\exp(\lambda[e^t - 1]) = \exp(\lambda[e^t - 1]) \cdot \lambda e^t$

$M'(t=0) = \lambda = E(x)$

$M''(t=0) = \lambda + \lambda^2 = E(x^2)$

Example 1.12:

$$X \sim Unif(0,1) \Rightarrow f(x) = \begin{cases} \frac{1}{1-0}, & 0 \le x \le 1 \\ 0, & e/w \end{cases}$$

Note: if $X \sim Unif(a,b)$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & e/w \end{cases}$$

$$E(X) = \mu = \frac{1}{2}(a+b) = \frac{1}{2}(0+1) = \frac{1}{2}$$

$$V(X) = \frac{1}{12}(b-a)^2 = \frac{1}{12}(1-0)^2 = \frac{1}{12}$$

$$M(t) = E(e^{tx}) = \int_0^1 e^{tx} \cdot 1 \, dx$$

$$= \frac{1}{t} e^{tx} \Big|_0^1$$

$$= t^{-1}e^t - t^{-1}$$

$$= t^{-1}(e^t - 1)$$

$$M'(t) = -t^{-2}(e^t - 1) + t^{-1}(e^t)$$

$$= -\frac{e^t - 1}{t^2} + \frac{e^t}{t}$$ * cannot divide by 0 *

$$M(t) = \frac{1}{t}(e^t - 1)$$

$$= \frac{1}{t}\left(\left[1 + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots\right] - 1\right)$$

$$= 1 + \frac{t}{2!} + \frac{t^2}{3!} + \frac{t^3}{4!} + \dots$$

$$M'(t) = 0 + \frac{1}{2} + \frac{2t}{3!} + \frac{3t^2}{4!} + \dots$$

$$\therefore E(x) = M'(t=0) = \frac{1}{2}$$

$$M''(t) = \frac{2}{3!} + \frac{6t}{4!} + \frac{12t^2}{5!} + \dots$$

$$\therefore E(x^2) = M''(t=0) = \frac{2}{3!} = \frac{2}{6} = \frac{1}{3}$$

$$V(x) = E(x^2) - E(x)^2$$

$$= \frac{1}{3} - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{3} - \frac{1}{4}$$

$$= \frac{1}{12}$$